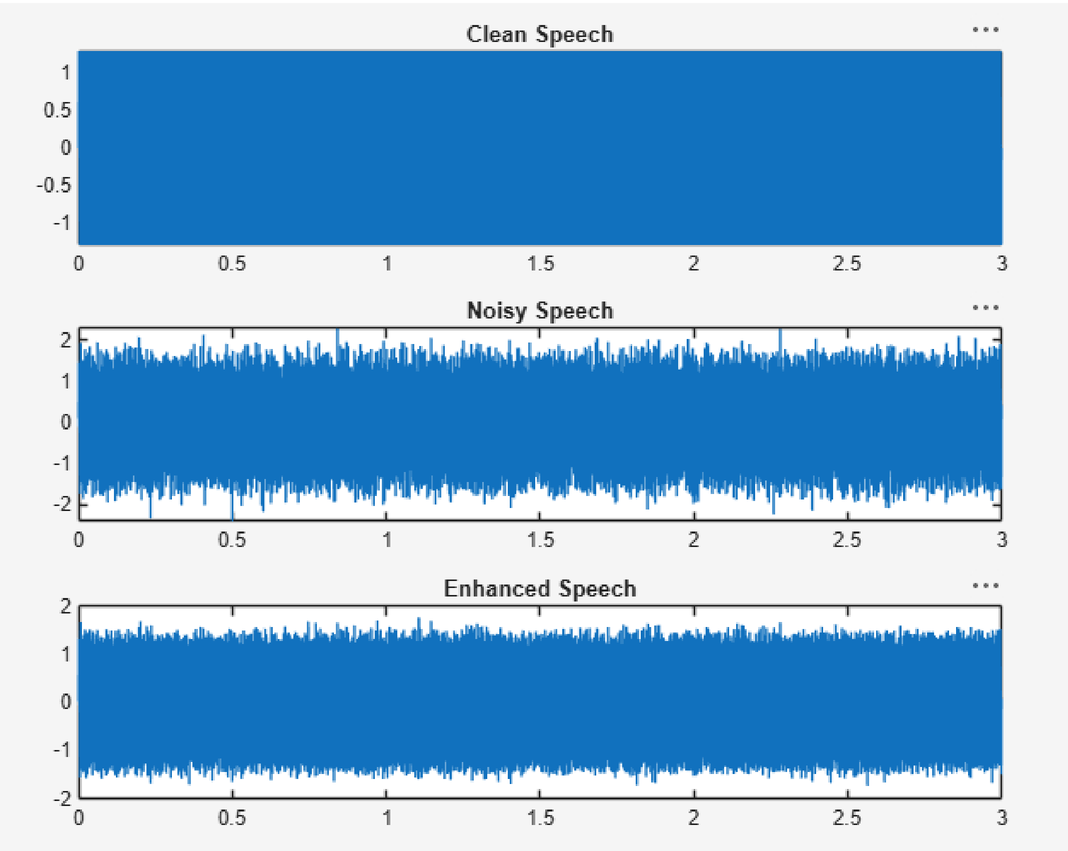
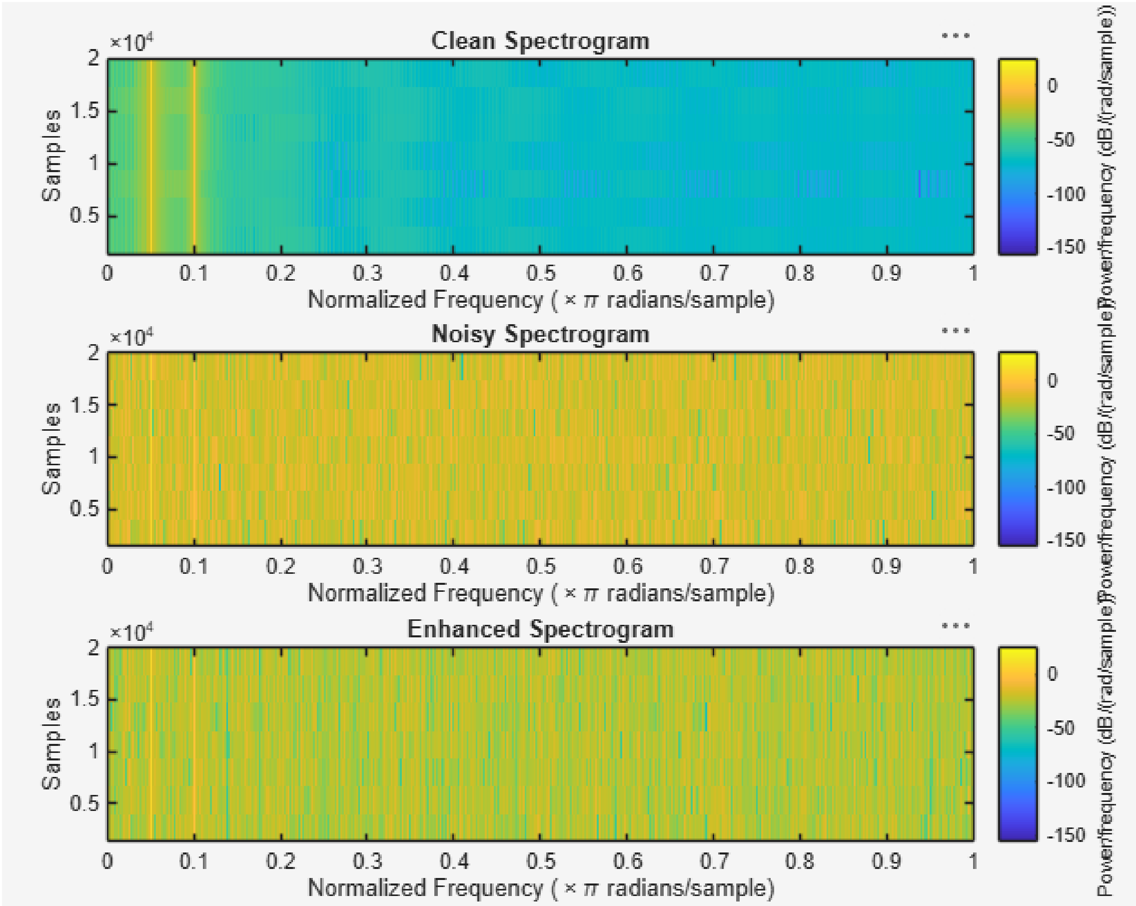
10. Implementation of Spectral Subtraction for Speech Noise Reduction

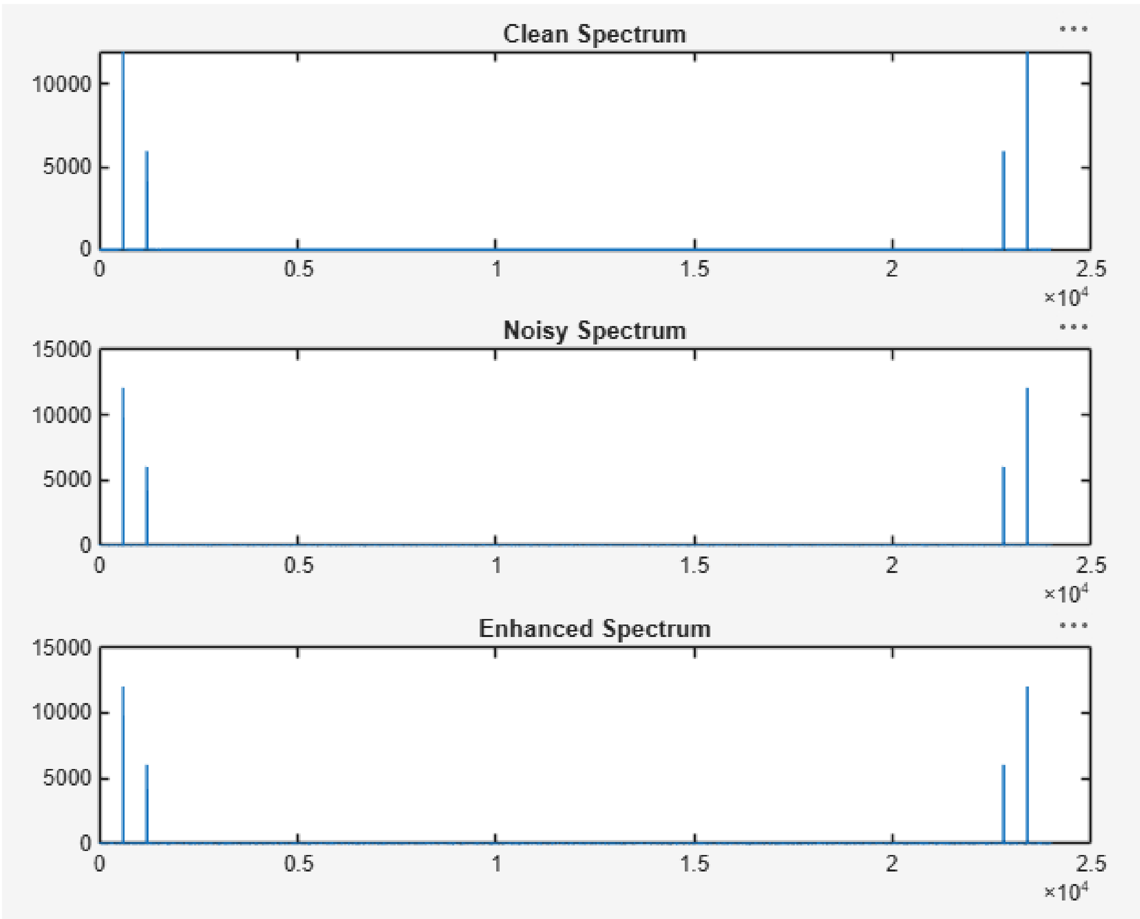
/MATLAB Drive/Dsp_10.m

```
clc; clear; close all;
Fs=8000; t=0:1/Fs:3;
x=sin(2*pi*200*t)+0.5*sin(2*pi*400*t);   % Clean signal
xn=x+0.3*randn(size(x));                 % Noisy signal
Y=fft(xn);
N=fft(0.3*randn(size(x)));
xe=real(ifft(max(abs(Y)-abs(N),0).*exp(1j*angle(Y))));
% Signals
figure;
subplot(311),plot(t,x),title('Clean Speech')
subplot(312),plot(t,xn),title('Noisy Speech')
subplot(313),plot(t,xe),title('Enhanced Speech')
% Spectrograms
figure;
subplot(311),spectrogram(x),title('Clean Spectrogram')
subplot(312),spectrogram(xn),title('Noisy Spectrogram')
subplot(313),spectrogram(xe),title('Enhanced Spectrogram')
% Magnitude Spectra
figure;
subplot(311),plot(abs(fft(x))),title('Clean Spectrum')
subplot(312),plot(abs(fft(xn))),title('Noisy Spectrum')
subplot(313),plot(abs(fft(xe))),title('Enhanced Spectrum')
```

Output:




Clean Spectrogram
Noisy Spectrogram
Enhanced Spectrogram
Samples
Normalized Frequency (× π radians/sample)
Power/frequency (dB/(rad/sample))


Clean Spectrum
Noisy Spectrum
Enhanced Spectrum